higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T430**(E)**(A14)T

# NATIONAL CERTIFICATE

# CONTROL SYSTEMS N6

(8080016)

**14 August 2019 (X-Paper)**
**09:00–12:00**

**REQUIREMENTS: 3-cycle semi-logarithmic graph paper**

**This question paper consists of 7 pages, a formula sheet of 2 pages, a Laplace transform table of 3 pages and 1 Nichols chart.**

Copyright reserved

# DEPARTMENT OF HIGHER EDUCATION AND TRAINING
# REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
CONTROL SYSTEMS N6
TIME: 3 HOURS
MARKS: 100

---

**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use neat, labelled circuit diagrams to answer questions when requested.
5. Use only BLUE or BLACK ink.
6. Use drawing equipment and pencil for ALL sketches and diagrams.
7. Work neatly.

---

Copyright reserved 

## QUESTION 1

Choose a term from COLUMN B to complete the description in COLUMN A. Write only the letter (A–K) next to the question number (1.1–1.10) in the ANSWER BOOK.

| | COLUMN A | | COLUMN B |
|---|---|---|---|
| 1.1 | A … is a graph drawn on an axis system containing an imaginary axis and a true axis from pole values for the open loop to determine the behaviour of the closed-loop system. | A | comparator |
| 1.2 | A… is a device which produces the acting signal which is the difference between the reference input and the feedback signal. | B | derivative control |
| 1.3 | … means the response is sluggish and oscillations never occur. | C | sine function |
| 1.4 | … means the correction made is proportional to the time derivative of the error. | D | overdamping |
| 1.5 | … is the point at which the magnitude graph cuts the zero $\alpha\beta$ line. | E | offset |
| 1.6 | … describes a system that is unstable when the output grows without any input. | F | set point |
| 1.7 | … is a function of time which changes periodically in a sinusoidal fashion at a frequency of ω rad/s and may be leading or lagging. | G | instability |
| 1.8 | … is the reference level associated with the decimal output of a system. | H | root locus |
| 1.9 | … means the response is as quick as possible without overshooting or oscillations. | I | critical damping |
| 1.10 | … is the difference between the equilibrium temperature and the desired value. | J | phase margin |
| | | K | proportional control |

(10 × 1) **[10]**

## QUESTION 2

Determine, with the aid of block diagram algebra reduction, the control ratios at both C1 and C2 in the block diagram in FIGURE 1.

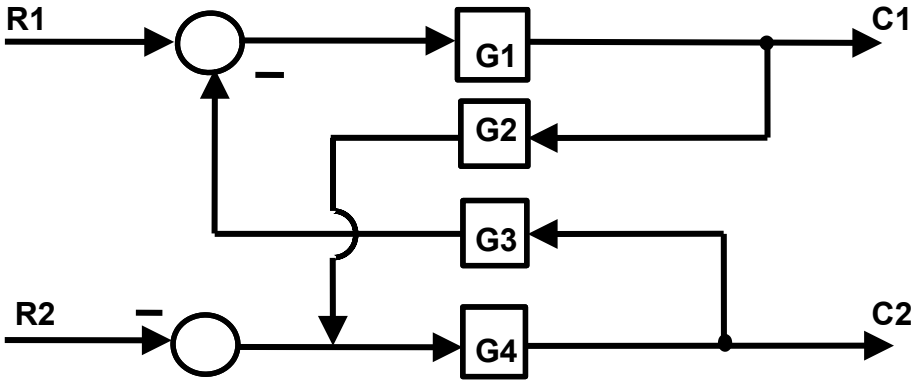



**FIGURE 1** **[10]**

## QUESTION 3

The transfer function of an open-loop control system is given below.

$$G(s)H(s)=\frac{35}{s(1+0{,}4\,s)}$$

Use the frequencies ω = 0,1; 1; 10 ; 50 ; 100 rad/s to answer the questions.

3.1 Calculate the log magnitude and phase values for the given frequencies in a table. (7)

3.2 Draw the Bode plot for the system on 3-cycle semi-logarithmic graph paper. (3)

**[10]**

## QUESTION 4

The attached Nichols chart illustrates a closed-loop gain versus phase plot.

4.1 Use the Nichols chart to determine each of the following:

4.1.1 Gain margin (1)

4.1.2 Phase margin (1)

4.1.3 Gain crossover frequency (1)

4.1.4 Phase crossover frequency (1)

4.1.5 Undamped natural resonant frequency (1)

Copyright reserved 

4.1.6 Peak frequency response (1)

4.1.7 Peak magnitude and phase (2)

4.1.8 Closed-loop bandwidth (1)

4.2 State whether the system is STABLE or UNSTABLE. (1)
**[10]**

## QUESTION 5

Use the transfer function for a root locus given below to determine each of the following:

$$G(s)H(s) = \frac{20(0{,}25s+1)}{s(0{,}05s+1)(0{,}5s+1)}$$

5.1 Open-loop sensitivity constant $K_o$ (3)

5.2 Poles and zeros of the function (2)

5.3 Centre of asymptotes on the real axis (2)

5.4 Asymptotic angles (3)
**[10]**

## QUESTION 6

6.1 Convert the Laplace transform functions below to a function of s:

6.1.1 $F(t) = 15\cos 135t$ (3)

6.1.2 $F(t) = \frac{60}{5}(1 - e^{-5t})$ (2)

6.2 Convert the Laplace transform functions to a function of t:

$$F(s) = \frac{45\omega}{(s^2 + \omega^2)(s+7)(s+5)}$$ (5)
**[10]**

Copyright reserved 

## QUESTION 7

A certain damped circuit has a damping ratio ($\zeta$) of 0,12 and an undamped natural frequency ($\omega_n$) of 6 rad/s.

Calculate each of the following:

7.1 Sine wave delay ($\Psi$) in radians

7.2 Amplitude $(\varphi)$ of the first peak

7.3 Percentage overshoot

7.4 Damping frequency $(\omega_d)$

7.5 Peak time for wave to reach maximum overshoot $(t_p)$

(5 × 2) **[10]**

## QUESTION 8

8.1 Draw a neat, labelled schematic diagram of the solid-state speed control of a series motor. (5)

8.2 Explain the operating principle of the solid-state speed control in QUESTION 8. (5)

**[10]**

## QUESTION 9

9.1 State the function of a pump. (1)

9.2 FIGURE 2 shows the operation of an external gear pump.

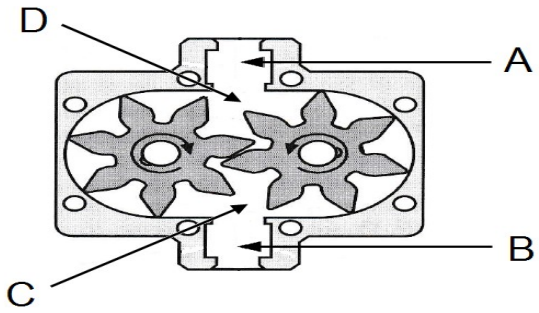



**FIGURE 2**

9.2.1 Name the parts labelled A to D. (4)

9.2.2 State the category under which this pump is classified. (1)

9.3 Explain the operation of an axial piston-type pump. (4)

**[10]**

## QUESTION 10

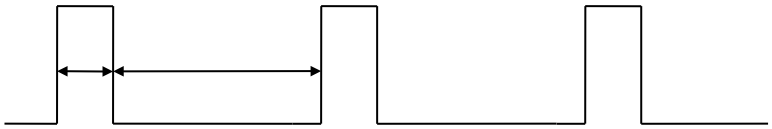
10.1 State the procedure for measuring the phase difference between TWO sinusoidal signals. (3)

10.2 Draw a neat, labelled diagram of a setup for testing the response of a system. (5)

10.3 A pulse train produced by a pulse generator has a mark-to-space ratio of 1:7.

Calculate the pulse repetition frequency of the waveform.

**HINT:** $prf = \frac{1}{Period}$



(2)
**[10]**

**TOTAL: 100**

Copyright reserved

**CONTROL SYSTEMS N6**

**FORMULA SHEET**

$\omega_d = \omega_n\sqrt{1-\zeta^2} = 2\pi f$ Peak time $t_p = \dfrac{\pi}{\omega_n\sqrt{1-\zeta^2}}$ $t_p = \dfrac{1}{f}$

*Number of oscillations* $\dfrac{t_s}{t_p}$ $\vee$ / *of* $\dfrac{2\sqrt{1-\zeta^2}}{\pi.\ \zeta}$

*Damping coefficient* $(\alpha) = \zeta\ .\omega_n = \dfrac{1}{\pi}\ \tau$

*Overshoot* $= e^{\frac{-\zeta\ \pi N}{\sqrt{1-\zeta^2}}}$

$\Psi = \dfrac{1}{\sqrt{1-\zeta^2}} \times \tan^{-1}\left[\dfrac{\sqrt{1-\zeta^2}}{-\zeta}\right] + \pi\ rad$

$Amplitude = \varphi = \left[1 + e^{\frac{-\zeta\pi N}{\sqrt{1-\zeta^2}}}\right]$

$\omega_n = \sqrt{\dfrac{Ko}{\tau}}$ $\tau + \dfrac{t_s}{4} = \dfrac{1}{\zeta\ .\ \omega_n}$

$\omega_p = \omega_n\ \sqrt{1-2\ \zeta^2}$

$\omega_b = \omega_n\ \sqrt{1-2\zeta^2 + \sqrt{2-4\zeta^2\ +4\zeta^4}}$

$\dfrac{C}{R} = \dfrac{G}{1\ \pm\ GH}$ $Sc = \dfrac{\sum P\ -\ \sum Z}{NP-NZ}$

$\zeta = \cos\ \varphi$ $\Psi = \dfrac{(2\,Ko\ +1)180°}{NP-NZ}$

$Ko = \dfrac{\Delta P_1\ .\ \Delta P_{2....}}{\Delta Z_1\ .\ \Delta Z_{2....}}$

Copyright reserved 

## AMPLIFIERS

$$Vo = - V_i \frac{R_f}{R_1}$$

$$Vo = - R_f \left[ \frac{V_1}{R_1} + \frac{V_2}{R_2} + \ldots \frac{V_N}{R_N} \right]$$

$$Vo = - RC \frac{d V_i(t)}{dt}$$

$$i_e = \frac{V_e}{R_e}$$

$$R_c = \frac{V_c}{i_c}$$

$$gm R_L = \frac{hfe}{hie} . R_L$$

$$t = \frac{1}{f}$$

## BODE AND NICHOLS CHARTS

$$Gain = 20 \log \left[ \frac{output}{input} \right] db$$

$$Phase = \sin^{-1} \left[ \frac{phase\ \ shift}{input} \right] - 180°$$

$$\tau = R . C$$

Copyright reserved

# LAPLACE TRANSFORM TABLE

| No | F(s) | f(t) |
|---|---|---|
| 1. | $1$ | $\delta(t)$ |
| 2. | $\frac{A}{s}$ | $A(t)$ $\{0 \quad t<0\}$ $\{A \quad t \geq 0\}$ |
| 3. | $\frac{1}{s}$ | $U(t)$ $\{0 \quad t<0\}$ $\{1 \quad t \geq 0\}$ |
| 4. | $\frac{A}{s^2}$ | $At$ |
| 5. | $\frac{2A}{s^3}$ | $At^3$ |
| 6. | $\frac{A_\omega}{s^2 + \omega^2}$ | $A\sin\omega t$ |
| 7. | $\frac{A_s}{s^2 + \omega^2}$ | $A\cos\omega t$ |
| 8(a). | $\frac{A}{\tau s + 1}$ | $\frac{A}{\tau} e^{\frac{-t}{\tau}}$ |
| 8(b). | $\frac{A}{s + a}$ | $Ae^{-at}$ |
| 9(a). | $\frac{A}{(\tau_1 s+1)(\tau_2 s+1)}$ | $\frac{A}{\tau_1-\tau_2} \left[e^{\frac{-t}{\tau_1}} - \frac{-t}{\tau_2}\right]$ |
| 9(b). | $\frac{A}{(s+a)(s+b)}$ | $\frac{A}{(b-a)} \left[e^{-at} - e^{-bt}\right]$ |
| 10(a). | $\frac{A}{(\tau_s+1)^2}$ | $\frac{At}{\tau^2} \left[e^{\frac{-t}{\tau}}\right]$ |
| 10(b). | $\frac{A}{(s+a)^2}$ | $Ate^{-at}$ |
| 11. | $\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ | $\frac{A\omega_n e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin\left(\omega_n\sqrt{1-\zeta^2}t\right.$ |
| 12(a). | $\frac{A}{s(\tau s+1)}$ | $A\left(1 - e^{\frac{-t}{\tau}}\right)$ |
| 12(b). | $\frac{A}{s(s+a)}$ | $\frac{A}{a}\left(1 - e^{-at}\right)$ |
| 13(a). | $\frac{A}{s^2(\tau s+1)}$ | $A\tau\left[e^{\frac{-t}{\tau}} + \frac{t}{\tau} - 1\right]$ |
| 13(b). | $\frac{A}{s^2(s+a)}$ | $\frac{A}{a^2}\left(e^{-at} + at - 1\right)$ |

Copyright reserved

| | | |
|---|---|---|
| 14(a). | $\frac{A\omega}{(s^2+\omega^2)(\tau s+1)}$ | $\frac{A\omega\tau}{1+\omega^2\tau^2}e^{\frac{-t}{\tau}}+\frac{A}{\sqrt{1+\omega^2\tau^2}}\sin(\omega t-\Psi)$ where $\Psi=\tan^{-1}\omega\tau \quad (0<\Psi<\pi)$ |
| 14(b). | $\frac{A\omega}{(s^2+\omega^2)(s+a)}$ | $\frac{A\omega e^{-at}}{(\omega^2+a^2)}+\frac{A}{\sqrt{\omega^2+a^2}}\sin(\omega t-\Psi)$ where $\Psi=\tan^{-1}\frac{\omega}{a} \quad (0<\Psi<\pi)$ |
| 15(a). | $\frac{A}{s(\tau_1 s+a)(\tau_2 s+1)}$ | $A\left[1+\frac{\tau_1 e^{\frac{-t}{\tau_1}}-\tau_2 e^{\frac{-t}{\tau_2}}}{\tau_1-\tau_2}\right]$ |
| 15(b). | $\frac{A}{s(s+a)(s+b)}$ | $\frac{A}{ab}\left[1+\frac{ae^{-bt}-be^{-at}}{b-a}\right]$ |
| 16(a). | $\frac{A}{s(\tau+a)^2}$ | $A\left[1-\frac{(\tau+t)}{\tau}e^{\frac{-t}{\tau}}\right]$ |
| 16(b). | $\frac{A}{s(s+1)^2}$ | $\frac{A}{a^2}\left[1-(1+at)e^{-at}\right]$ |
| 17. | $\frac{A\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$ | $A\left[1+\frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_n t\sqrt{1-\zeta^2}-\Psi\right]$ where $\Psi=\tan^{-1}\frac{\sqrt{1-\zeta^2}}{-\zeta}$ $(0<\Psi<\pi)$ |
| 18(a). | $\frac{A}{s^2(\tau_1 s+1)(\tau_2 s+1)}$ | $A\left[t-\tau_1-\tau_2-\frac{\tau_2^2 e^{\frac{-t}{\tau_2}}-\tau_1^2 e^{\frac{-t}{\tau_1}}}{\tau_1-\tau_2}\right]$ |
| 18(b). | $\frac{A}{s^2(s+a)(s+b)}$ | $\frac{A}{ab}\left[t-\frac{a+b}{ab}-\frac{\frac{b}{a}e^{-bt}-\frac{a}{b}e^{-at}}{b-a}\right]$ |
| 19(a). | $\frac{A}{s^2(\tau s+1)^2}$ | $A\left[t-2\tau+(t+2\tau)e^{\frac{-t}{\tau}}\right]$ |
| 19(b). | $\frac{A}{s^2(s+a)^2}$ | $\frac{A}{a^2}\left[t-\frac{2}{a}+\left(t+\frac{2}{a}\right)e^{-at}\right]$ |
| 20. | $\frac{A\omega_n^2}{s^2(s^2+2\zeta\omega_n s+\omega_n^2)}$ | $A\left[\tau-\frac{2\zeta}{\omega_n}+\frac{e^{-\zeta\omega_n t}}{\omega_n\sqrt{1-\zeta^2}}\sin(\omega_n t\sqrt{1-\zeta^2}-\Psi\right]$ where $\Psi=2\tan^{-1}\frac{\sqrt{1-\zeta^2}}{-\zeta}$ $(0<\Psi<\pi)$ |

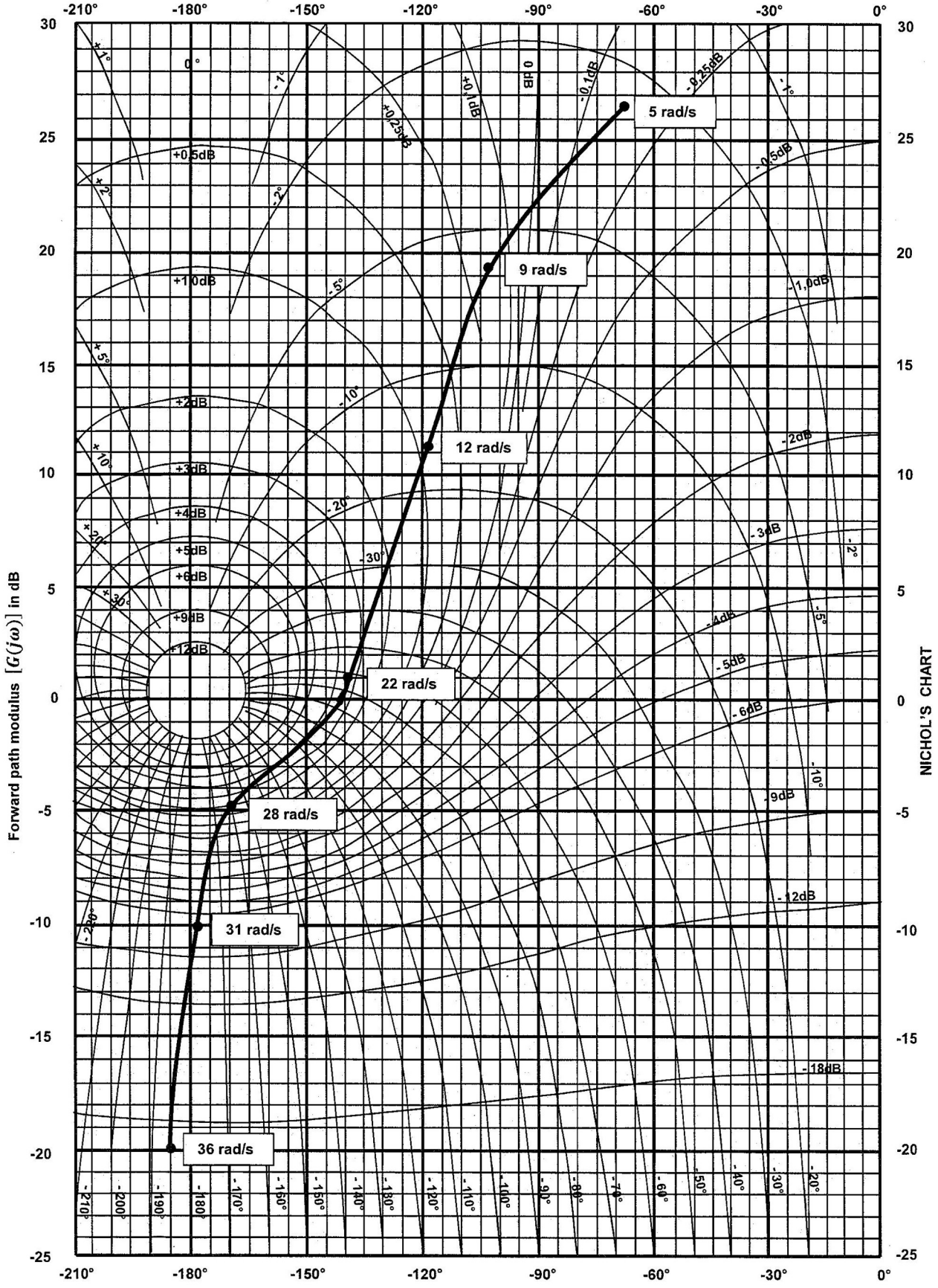
Copyright reserved 

| | | |
|---|---|---|
| 21(a). | $\frac{A\omega}{(s^2+\omega^2)(\tau_1 s+1)(\tau_2 s+1)}$ | $A\left[\frac{\tau_1^2\omega e^{\frac{-t}{\tau_1}}}{(\tau_1-\tau_2)(1+\omega^2\tau_1^2)}+\frac{\tau_2^2\omega e^{\frac{-t}{\tau_2}}}{(\tau_1-\tau_2)(1+\omega^2\tau_2^2)}+\frac{}{(1+\omega$<br>*where*<br>$\Psi=\tan^{-1}\omega\tau_1+\tan^{-1}\omega\tau_2$ |
| 21(b). | $\frac{A\omega}{(s^2+\omega^2)(s+a)(s+b)}$ | $A\left[\frac{\omega e^{-at}}{(b-a)(\omega^2+a^2)}+\frac{\omega e^{-bt}}{(a-b)(\omega^2+b^2)}+\frac{\sin(\omega t-$<br>$(\omega^2+a^2)(\omega$<br>*where*<br>$\Psi=\tan^{-1}\frac{\omega(a+b)}{ab-\omega^2}$ (0 < Ψ < π) |
| 22(a). | $\frac{A\omega}{(s^2+\omega^2)(\tau s+1)^2}$ | $\frac{A}{1+\omega^2\tau^2}\left[\frac{\omega t+2\omega\tau}{1+\omega^2\tau^2}e^{\frac{-t}{\tau}}+\sin(\omega t-\Psi\right]$<br>*where*<br>$\Psi=2\tan^{-1}\omega\tau$ |
| 22(b). | $\frac{A\omega}{(s^2+\omega^2)(s+a)^2}$ | $\frac{A}{\omega^2+a^2}\left[\frac{a\omega(at+2)e^{-at}}{\omega^2+a^2}+\sin(\omega t-\Psi\right]$ |
| 23. | $\frac{A\omega\omega_n^2}{(s^2+\omega^2)(s^2+2\zeta\omega_n s+\omega_n^2)}$ | $\frac{A\omega_n^2}{[(\omega_n^2-\omega^2)^2+4\zeta^2\omega^2\omega_n^2]^{\frac{1}{2}}}$<br>$\left[\sin(\omega t-\Psi)+\frac{\omega e^{-\zeta\omega_n t}\sin(\omega_n\sqrt{1-\zeta^2}t-\Psi_2)}{\omega_n\sqrt{1-\zeta^2}}\right]$<br>*where*<br>$\Psi_1=\tan^{-1}\left[\frac{2\zeta\omega\omega_n}{\omega_n^2+\omega^2}\right]\ (0<\Psi_1<\pi)$<br>and<br>$\Psi_2=\tan^{-1}-\frac{2\zeta\omega_n^2\sqrt{1-\zeta^2}}{\omega^2-\omega_n^2(1-2\zeta^2)}\ (0<\Psi_2<\pi)$ |

Copyright reserved

(8080016)

## NICHOLS CHART




Copyright reserved